

E-Commerce Dataset — Data Description

Five CSV files/tables are used. The first three contain raw Olist data, while the last two contain monthly aggregated/derived data.

1. olist_orders_dataset.csv

Purpose: Order-level information, including order status and different order/delivery dates.

Main key/relationship: order_id

Column	Description
order_id	Unique identifier of the order
customer_id	Unique identifier of the customer who placed the order
order_status	Current status of the order, such as delivered, cancelled, unavailable, etc.
order_purchase_timestamp	Date and time when the customer placed the order
order_approved_at	Date and time when the order/payment was approved
order_delivered_carrier_date	Date and time when the order was handed over to the carrier
order_delivered_customer_date	Actual date and time when the customer received the order
order_estimated_delivery_date	Estimated date by which the order was expected to be delivered

2. olist_order_items_dataset.csv

Purpose: Item-level information for each order, including product, seller, price and freight.

Main key/relationship: order_id → olist_orders

Column	Description
order_id	ID of the order to which the item belongs
order_item_id	Sequential identifier of an item within an order
product_id	Unique identifier of the product
seller_id	Unique identifier of the seller
shipping_limit_date	Date by which the seller should ship the item
price	Selling price of the item
freight_value	Freight/shipping charge associated with the item

3. olist_order_reviews_dataset.csv

Purpose: Customer review and feedback information. Contains structured ratings and unstructured customer comments.

Main key/relationship: order_id → olist_orders

Column	Description
review_id	Unique identifier of the review
order_id	ID of the order being reviewed
review_score	Customer rating/score, generally from 1 to 5
review_comment_title	Title of the customer's review

Column	Description
review_comment_message	Text/message written by the customer
review_creation_date	Date and time when the review was created
review_answer_timestamp	Date and time when the review was answered

4. olist_monthly_real_metrics.csv

Purpose: Monthly aggregated metrics calculated from the real Olist data. Data type: Derived from real Olist data.

Main key/relationship: month

Column	Description
month	Month used for monthly aggregation
orders	Total number of orders during the month
order_items	Total number of order items sold
revenue	Total sales/revenue during the month
avg_order_value	Average revenue/value per order
avg_item_price	Average selling price of items
total_freight	Total freight value
avg_delivery_days	Average actual delivery time in days
median_delivery_days	Median actual delivery time in days
avg_estimated_days	Average estimated delivery time
return_count	Number of returns
return_rate	Return percentage/rate
products_sold	Number of products/items sold
active_sellers	Number of sellers active during the month
cancelled_count	Number of cancelled orders
kw_delivery_count	Number of reviews/comments identified as delivery-related
kw_product_count	Number of reviews/comments identified as product-related
kw_price_count	Number of reviews/comments identified as price-related
kw_service_count	Number of reviews/comments identified as service-related
kw_avail_count	Number of reviews/comments identified as availability-related

5. synthetic_monthly_business_context.csv

Purpose: Monthly synthetic business-context data for variables not available in the original Olist dataset. Data type: Synthetic.

Main key/relationship: month

Column	Description
month	Month corresponding to the Olist monthly timeline
inventory_gap_pct	Estimated percentage of inventory affected by stock/inventory gaps
product_availability_pct	Estimated percentage of products available
competitor_pressure_index	Index representing the level of competitive pressure
competitor_discount_pct	Estimated average competitor discount percentage
competitor_price_index	Index representing competitor pricing relative to the business
market_demand_index	Index representing overall market demand
ad_spend	Estimated monthly advertising/marketing expenditure

Column	Description
website_visits	Estimated monthly website visits
website_conversion_rate	Estimated percentage of website visitors who convert into orders
promotion_index	Index representing the intensity of promotional activity
marketing_campaign_count	Number of marketing campaigns during the month
customer_acquisition_cost	Estimated cost of acquiring one customer
support_ticket_volume	Estimated number of customer support tickets
new_product_launch_count	Number of new products launched during the month
shipping_cost_index	Index representing shipping-cost conditions
stockout_event_count	Number of inventory stockout events

Quick Table Summary

CSV / Table	Data Level	Main Information	Data Type
olist_orders_dataset.csv	Order-level	Orders, status, purchase & delivery dates	Real
olist_order_items_dataset.csv	Item-level	Products, sellers, price, freight	Real
olist_order_reviews_dataset.csv	Review-level	Rating, review title & customer comments	Real
olist_monthly_real_metrics.csv	Month-level	Sales, orders, delivery, reviews, returns, sellers etc.	Derived from real data
synthetic_monthly_business_context.csv	Month-level	Inventory, competition, marketing, traffic, promotions, market conditions etc.	Synthetic

Main Relationships

olist_orders → **olist_order_items** using **order_id**

olist_orders → **olist_order_reviews** using **order_id**

olist_monthly_real_metrics ↔ **synthetic_monthly_business_context** using **month**